

Measuring Consumer Preferences for Used Battery Electric Vehicles

Zain Hoda¹, Xiatian Iogansen¹, John Helveston¹

¹Department of Engineering Management and Systems Engineering, George Washington University, Washington, D.C., USA

1 Motivation and Research Questions

Used vehicles account for approximately 70% of all vehicle transactions in the United States [1], yet the used battery electric vehicle (BEV) segment has received comparatively little systematic inquiry from researchers or policymakers. This inattention is increasingly difficult to justify: used EV sales rose 34.2% year-over-year as of February 2025, while average days' supply declined 21.5% over the same period [2], signaling demand growth that is outpacing inventory. The used BEV market also carries significance beyond its transaction volume. It extends the productive life of vehicles and battery packs, supports circular economy objectives [3], and reduces the effective entry price for lower- and middle-income households seeking the operating cost advantages of electric propulsion without the capital outlay of a new vehicle purchase.

Consumer demand for used BEVs remains constrained, however, by barriers that do not apply equally to conventional used vehicles. Survey evidence indicates that 52% of U.S. adults are unlikely to consider a used EV purchase, compared with 28% who express openness to doing so [4]. Used BEVs share certain adoption barriers with new EVs — range anxiety, charging infrastructure availability — but carry additional purchase risks specific to the secondary market. Among these, battery condition is the most consequential: it is the primary determinant of a used BEV's residual range, longevity, and resale value, yet it is largely unobservable to prospective buyers at the point of sale. Unlike mileage or model year, which serve as widely accepted proxies for mechanical wear in conventional vehicles, battery degradation trajectories vary substantially among BEVs of identical vintage and odometer reading as a function of prior charging behavior, thermal exposure, and duty cycle [5]. This structural information asymmetry creates conditions under which buyers systematically discount older BEVs as a hedge against unobserved deterioration, potentially suppressing market prices below the true value of well-maintained vehicles and dampening demand across the segment.

This study addresses three interrelated research questions:

1. What is consumers' willingness to pay (WTP) for a used BEV powertrain relative to a comparable internal combustion engine (ICE) vehicle, and how does this WTP vary across price budget tiers?

2. How do the principal vehicle attributes — driving range, vehicle age, and powertrain type — jointly determine WTP for used BEVs?
 3. To what extent do preference structures and WTP magnitudes differ between consumers who self-identify as likely versus unlikely future BEV adopters?
-

2 Background and Contribution

The EV adoption literature is dominated by studies of new vehicle buyers, in which purchase price, operating costs, driving range, recharging time, and charging infrastructure availability are consistently identified as the primary determinants of choice [6]. Relatively little of this work extends to the used-vehicle segment, which serves a distinct population with different cost constraints, vehicle quality expectations, and information environments. Affordability, vehicle condition, and reliability are the dominant evaluation criteria for used-vehicle buyers [6]; total cost of ownership and fuel costs are also salient, particularly among cost-constrained households. Average used-vehicle transaction prices have remained elevated since the post-pandemic period, reaching \$25,547 in April 2025 [7], compressing the potential cost advantage of used BEVs for budget-sensitive buyers.

Among the limited studies that do examine the used EV market specifically, qualitative and descriptive analyses predominate. Segmentation work identifies three broadly distinct prospective buyer types: those for whom purchase cost constitutes a binding constraint, those resistant to the behavioral and technological adjustments associated with electric propulsion, and those motivated by a favorable cost-to-benefit ratio relative to accelerated new EV depreciation [8]. The demographic profile of observed used EV owners — disproportionately male, middle-aged, higher-income, and highly educated — closely mirrors that of new EV owners [9], indicating that the secondary market has not yet substantially extended access to lower-income households. Pre-purchase concerns center on battery condition, charging access, and vehicle price; lower-income prospective buyers weight both price and charging availability more heavily than higher-income counterparts [10].

This literature, while informative about the qualitative structure of buyer heterogeneity, has not produced disaggregated quantitative estimates of WTP for powertrain type in the used BEV market. The present study addresses that gap through a discrete choice experiment (DCE) administered to a nationally distributed sample of prospective used-vehicle buyers in the United States. The DCE design recovers implicit monetary valuations for vehicle attributes directly from observed choices between competing alternatives, enabling WTP estimation disaggregated by price budget tier, adoption propensity, vehicle type (car vs. SUV), driving range, and vehicle age. To the authors’ knowledge, this constitutes the first systematic quantification of used BEV willingness to pay at this level of segmentation in a U.S. context.

3 Data and Methods

3.1 Survey Design

The survey targets U.S. residents aged 18 years or older who report an intention to purchase a used car or SUV within the following two years. The core instrument is a DCE in which respondents complete six repeated choice tasks. Each task presents three used vehicle alternatives — differentiated on observed attributes — alongside a “none of the above” opt-out option. Respondents select their preferred alternative in each task. This repeated-measures design exploits within-respondent variation across attribute combinations to estimate preference parameters more precisely than a single-task design would permit.

Vehicle attributes included in the DCE are mileage, powertrain type, operating cost, model year, and purchase price. Powertrain is defined at three levels: conventional internal combustion engine vehicle (ICEV), gas hybrid electric vehicle (HEV), and battery electric vehicle (BEV), with the ICEV serving as the reference category throughout. Operating cost is expressed in cents per mile driven. Purchase price levels are segmented by vehicle type and consumer budget tier to maintain contextual plausibility of the attribute ranges: low-budget cars (\$10,000–\$20,000), high-budget cars (\$20,000–\$40,000), low-budget SUVs (\$15,000–\$25,000), and high-budget SUVs (\$25,000–\$45,000). All attribute definitions are presented to respondents before the choice tasks begin to ensure consistent interpretation across the sample.

In addition to the DCE, the survey collects information on respondents’ current vehicle ownership and primary vehicle characteristics, future purchase intentions and anticipated budget, EV-specific knowledge and attitudes (assessed through 5-point Likert scales covering range anxiety, charging infrastructure concerns, environmental beliefs, cost perceptions, and battery-related concerns), and standard sociodemographic characteristics including household income, housing type, education level, and employment status.

3.2 Data Collection and Sample

The survey was administered online between December 2025 and April 2026 across two established research platforms. Respondents who failed embedded attention checks, exhibited within-survey response inconsistencies, or provided uninformative filler text in open-ended fields were removed from analysis. The final analytic sample comprises **2,916 respondents** distributed across the continental United States. Median survey completion time was approximately 14 minutes.

3.3 Estimation

Consumer choice is modeled within the random utility framework. Each respondent i is assumed to select the alternative j that maximizes their latent utility U_{ij} in each choice task,

where utility is a function of observed vehicle attributes and an unobserved idiosyncratic error term assumed to follow an independent Gumbel distribution.

A multinomial logit (MNL) model, estimated via maximum likelihood, provides the baseline specification under the assumption of preference homogeneity across respondents. To relax this restriction — which is inconsistent with the substantial heterogeneity in EV attitudes documented in the literature — the primary analysis employs a mixed logit (MXL) model. In the MXL specification, each of the eight preference parameters is treated as a draw from an independent normal mixing distribution:

$$\beta_{ik} \sim \mathcal{N}(\mu_k, \sigma_k^2)$$

where μ_k is the population mean valuation for attribute k and σ_k captures the dispersion of that valuation across respondents. A statistically significant estimate of σ_k constitutes evidence that preferences for attribute k are genuinely heterogeneous in the population, rather than uniform. The eight parameters specified as randomly distributed are the BEV powertrain indicator, HEV powertrain indicator, BEV driving range, vehicle mileage, vehicle age, operating cost, purchase price, and the alternative-specific constant for the no-choice option.

Six MXL models are estimated, stratified by self-reported adoption propensity (likely vs. unlikely BEV adopter) and vehicle segment (full sample, car segment, SUV segment), yielding models for all combinations of these two dimensions. All models are estimated in R using the `logitr` package (v1.1.2) under a preference-space specification. The mixing integral is approximated using 100 Sobol quasi-random draws per respondent. To reduce sensitivity to starting values and mitigate convergence to local optima, each model is initialized from 20 independent random starting points; the run achieving the globally lowest log-likelihood is retained. Convergence is verified via the relative and absolute function tolerance criterion (Exit Status 3) in all six models.

Willingness to pay for attribute k is derived as the negative ratio of the mean attribute coefficient to the mean price coefficient:

$$\widehat{\text{WTP}}_k = -\frac{\hat{\mu}_k}{\hat{\mu}_{\text{price}}}$$

This quantity represents the dollar amount the average respondent would pay — or require as a compensating discount — for a one-unit change in attribute k , holding utility constant. Uncertainty bounds on WTP are computed by drawing 10,000 samples from the estimated joint distribution of $(\hat{\mu}_k, \hat{\mu}_{\text{price}})$ and reporting the resulting 95% simulation interval.